

Research Companion

A plain-English guide to every feature — and how to use them.

Local-first research assistant · turns your PDFs and arXiv links into a knowledge graph, a chat interface, and a full reviewer-style report. Source version 0.7.x. Legend: **DETERMINISTIC** no AI needed · **USES AI** needs a key ·

OPT-IN off until you turn it on.

The one-minute picture. You add papers (from an arXiv link, a DOI, a Semantic Scholar entry, or a local PDF). Research Companion reads each one, pulls out the ideas, methods, datasets, claims and results, and connects them into a single knowledge graph across all your papers. From there you can **explore** it visually, **chat** with it (every answer cites the paper it came from), and run a full **review** that tells you whether a draft is ready to submit and what to fix first. Everything runs on your machine; nothing is uploaded unless you explicitly turn on a feature that needs the internet.

1 Build your paper library

Get papers into the tool and turn them into a connected graph.

Add papers **DETERMINISTIC**

Pull in a paper from almost any source. arXiv links and IDs, DOIs, Semantic Scholar IDs, or a PDF already on your computer all work. Titles, authors and year are read automatically.

Use: `research-companion add https://arxiv.org/abs/2404.16130` · a DOI, an S2 ID, or `./my-paper.pdf` ·
add a whole list at once with `add -f papers.txt` .

Cost estimate before you build **DETERMINISTIC**

See roughly what reading your papers with an AI model will cost before you spend anything. Re-building later is free — results are cached and only re-run when needed.

Use: `research-companion cost-estimate`

Build the knowledge graph **USES AI**

Reads every paper and extracts its concepts, methods, datasets, claims, and results, then merges them across papers. When the same method appears in five papers it becomes one shared node, so you can see what your papers have in common at a glance.

Use: `research-companion build`

Better PDF reading (optional) **OPT-IN**

The basic reader handles normal digital PDFs. Installing the optional Docling engine adds OCR for scanned/image PDFs plus better handling of columns, tables and figures. Scanned PDFs with no text fail with a clear message instead of entering your library empty.

Use: `pip install research-companion[docling]` — auto-detected once installed.

2 Explore and understand your papers

Navigate the graph and ask it questions — with answers you can verify.

Interactive graph view DETERMINISTIC

Opens your whole library as a clickable graph in the browser — concepts, methods, datasets and the papers that mention them, with the connections between them.

Use: `research-companion view`

Chat with your library USES AI

Ask a question in plain language and get an answer built from your papers. Every fact cites the exact paper it came from, so you can check it before you trust it. It follows the connections in the graph rather than guessing.

Use: `research-companion chat "how do GraphRAG and LightRAG differ?"` · or just `chat` for an interactive session.

Search the graph DETERMINISTIC

Find a specific concept, method or dataset without opening the browser.

Use: `research-companion search "attention" --kind concept`

3 The Research Lab — a visual workspace

A browser workspace that grows your graph live as papers arrive. Needs the server extra: `pip install -e "[server]"`.

Live graph & folder ingest DETERMINISTIC

Point it at a folder of PDFs and watch each one appear in the graph in real time. You see every file's progress and can pick exactly which PDFs to bring in.

Use: `research-companion lab serve` then open the page · or `lab ingest <folder>`.

Draft alignment USES AI

Tell the Lab which paper is *your* draft, and it scores every other paper against it — does it strengthen, challenge, or offer an alternative to your work — backed by real evidence quotes, with papers colored by evidence strength.

Use: `research-companion set-draft <paper-id>` then `align <paper-id>`.

Ask & Compare USES AI

Ask gives section-scoped answers that cite the exact section they came from (cheaper and more precise than reading a whole paper). **Compare** does a structured head-to-head of any two papers.

Use: `research-companion ask "how does X compare to Y?"` · `compare <paper-a> <paper-b>`.

Citation coverage **DETERMINISTIC**

Reads your draft's bibliography and shows which cited papers you already have and which are missing — with a one-click "add all" — and always tells you honestly when an analysis is running on only part of your references.

Use: built into the Lab once you set a draft.

4 Get a paper submission-ready — the review team

One command runs a team of specialised checkers over a paper. The report opens with a single go/no-go verdict and a "fix this first" list.

Submission-readiness verdict. Every review starts with one honest answer — **ready**, **revise**, or **not ready** — plus a prioritized list of what to fix first, gathered from whatever checks actually ran. Things that would get a paper desk-rejected are "must fix"; everything else is a "should review" warning. If a check didn't run, the report says so plainly rather than pretending the paper passed it.

Run a review **USES AI** **DETERMINISTIC CORE**

Runs all the lanes below at once. Add `--venue neurips` to also check venue fit and desk-reject rules; add `--fast` to skip the AI lanes and run entirely offline; add `--report out/` for an HTML + JSON report, or `--serve` for a live dashboard.

Use: `research-companion review <paper-id> --venue neurips --report out/`

What each lane checks

- **Citations** — verifies every reference against public databases; flags wrong or unverifiable ones.
- **Prior art & novelty** — finds related work and checks whether the paper's claimed contributions are actually new, verifying every evidence quote against the paper's own text.
- **Statistical soundness** — recomputes reported p-values and sanity-checks reported averages; flags numbers that don't add up. **DETERMINISTIC**
- **Reproducibility** — looks for code/data links, availability statements, and reporting-checklist mentions → high / medium / low.
- **Ethics & integrity declarations** — checks for funding, conflict-of-interest, ethics/consent, and author-contribution statements.
- **Overlap** — flags passages that closely match another paper in your library.
- **Venue fit & compliance** (with `--venue`) — does it fit the venue's scope, and does it break any desk-reject rule (page limit, missing sections, anonymization leaks)?

Reviewer's take (AI narrative) **USES AI** **OPT-IN**

Turn this on and the report adds a short, friendly "reviewer's take" paragraph plus an ordered fix plan, written by the AI. It is grounded strictly in the verdict above — it can reword and reorder the fix list, but it cannot invent new problems or change the result. Off by default.

Use: turn on the `readiness_narrative` setting (Settings page, or the CLI settings).

5 Trust & verification tools

Small, sharp, deterministic checks you can run on their own — and a way to hand them to other AI agents.

Reference checker **DETERMINISTIC**

Validates a paper's whole bibliography against CrossRef, OpenAlex, arXiv and Semantic Scholar and marks each reference verified, suspect, or unverified.

Use: `research-companion refcheck <paper-id>`

Domain connectors: PubMed · Europe PMC · DBLP **OPT-IN**

Turn these on to verify references and search prior art in biomedical (PubMed, Europe PMC) and computer-science (DBLP) databases the general sources miss. Off by default; the tool behaves exactly as before until you enable them, and nothing extra is sent anywhere unless you do.

Use: add `--connectors europepmc,pubmed,dblp` to `refcheck` or `review`, or turn them on in Settings.

Desk-reject compliance linter **DETERMINISTIC**

Checks a paper against a venue's submission rules for the mistakes that get papers rejected before anyone reads them: over the page limit, a missing required section (e.g. Limitations), an anonymization leak in a double-blind submission, or an incomplete bibliography.

Use: `research-companion check-compliance <paper-id> --venue neurips`

Statistical soundness (Statcheck + GRIM) **DETERMINISTIC**

Re-derives each reported p-value from the test statistic and degrees of freedom and flags mismatches, and checks whether reported means are even arithmetically possible for the sample size. No AI — pure maths.

Use: `research-companion check-stats <paper-id>`

Near-duplicate & paraphrase overlap **DETERMINISTIC** **OPT-IN EXTRA**

Flags passages in a paper that closely match another paper already in your library — useful for catching self-plagiarism or a draft too close to prior work. By default it compares wording locally (nothing leaves your machine). An opt-in setting adds an on-device "meaning" comparison that also catches reworded or translated reuse.

Use: `research-companion check-overlap <paper-id>` · add `--semantic` for the paraphrase pass.

MCP trust-layer server **DETERMINISTIC**

Exposes four key-free verification tools — verify a citation, ground a claim in a source, check citation coverage, and search your library — so any MCP-capable AI assistant can call them and get trustworthy, checkable answers from your library.

Use: `research-companion mcp serve`

6 Discover what you're missing

Topic search & citation expansion DETERMINISTIC

Find papers you don't have yet. Search a topic (ranked by citation count), or follow the references and citations of the papers you already have to surface the important ones you're missing. No API key needed.

Use: `research-companion discover "graph-based RAG" · discover --expand · add --add` to pull them in.

7 Answer reviewers (rebuttal assistant)

Grounded rebuttal drafts USES AI

Give it your reviews and it drafts point-by-point replies that quote only real passages from your paper. Anything it can't back up with a real quote is flagged for you to check rather than shipped. Duplicate concerns from multiple reviewers are grouped, and a planned-revisions changelog is assembled for you.

Use: `research-companion rebuttal <paper-id> --reviews reviews.txt --tone firm`

8 Get your data in and out

Export the graph DETERMINISTIC

Export your knowledge graph to use elsewhere — as an Obsidian vault (fully linked notes), markdown, CSV (for spreadsheets or Neo4j), or raw JSON.

Use: `research-companion export --format obsidian --output ./vault/`

Reference-manager interoperability DETERMINISTIC

Export your references as BibTeX or RIS, import a `.bib` file from Zotero or Mendeley, or resolve the `\cite` keys in a LaTeX file against your library.

Use: `research-companion export-bib · import-bib refs.bib · cite-tex paper.tex ·`

9 How it keeps you in control

- **Local-first.** Your papers and graph live on your machine (in `~/.research-companion/`). Nothing is uploaded unless you use a feature that needs the internet.
- **Opt-in where it matters.** Anything that costs money (AI lanes), sends text to an outside service (remote connectors, the paraphrase pass's remote mode), or writes prose (the reviewer's take) is **off by default**. When a feature is off, the tool behaves exactly as if it didn't exist.
- **Honest by design.** Checks that can't run say so; answers cite their sources; the AI narrative can never invent a finding or overturn the deterministic verdict.
- **Your keys, your file.** API keys live in a local `.env` and are managed from the Settings page. Semantic search's `HF_TOKEN` is optional; without it, a keyword fallback is used.

